

QECCertificates: checkable distance certificates for quantum error correction

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Abstract

A quantum computer fails unless the code that stores its information is good enough, and one number measures how good that code is: the size of the smallest error it cannot detect. Quantum low-density parity-check codes promise a favourable trade between that number and the number of physical parts, and they are found by search, which enumerates candidates and computes a distance for each. In practice the public data report such a distance as an upper bound, because certifying it is expensive. A reader who wants to know whether the number is real has to trust the program, or repeat the search. Here we show that the check can be built from theorems instead of trust. The encoding that turns a code into a satisfiability instance is faithful in both directions, so a model of the encoded formula is exactly a light logical operator, and the checker that reads a solver’s refutation is sound. A distance claim can then be closed by a kernel that never saw the search. The closest previous route replays the solver’s proof inside a proof assistant, and its authors report that at 144 qubits the replay is the wall; separating search from check keeps the expensive part outside the trusted base. Fifty-eight Lean 4 modules carry the algebra, the translation from Pauli operators to bit-vectors and the code instances, and every non-private theorem and lemma is printed in an audit region whose statements rest on no axiom beyond the three the kernel’s logic needs. A certificate that is checkable without the tool that produced it can travel to other constructions and other solvers, and two companion developments already import the library as the home of their shared statements.

1 Why this exists

Quantum low-density parity-check codes [1] are designed by search: choose a construction, enumerate candidates, compute a distance. The public schema behind the qLDPC Challenge [2] records what a search leaves open. A distance that is not certified is reported there as an upper bound, for a non-CSS code because the Pauli-weight certifier “is not available yet” and for a circuit-level distance because an exact tier “is deferred future work”; what lies between an upper bound and a proof, the schema calls evidence.¹

¹The quotations are from the schema’s own descriptions of the distance tiers, read on 30 September 2026. The wording is theirs, not ours.

Two notions in that sentence name what is missing, and both need a word of explanation. A CSS code is one whose parity checks split into two families, one measuring X-type errors and one measuring Z-type; the Pauli-weight certifier is the component that would bound how few qubits a logical operator touches. A circuit-level distance counts faults in the syndrome extraction that reads those checks, not only faults in the stored codewords, and its code-level analogue is what this paper calls a fault distance.

The nearest prior work on the kernel side, Lean-QEC [3], turns the distance question into a satisfiability question and replays the solver’s proof inside a proof assistant. Its reported reach is 90 qubits, and its authors state that at 144 qubits the replay itself is the wall.

This paper reports the library that supplies both halves of the check. A distance claim becomes checkable when the search problem is encoded faithfully, so that solving the encoded formula and finding a light logical operator are the same thing, and when the refutation the solver returns is read by a checker whose soundness is a theorem. The search may then be as fast, as heuristic and as untrusted as its users like, because it is no longer part of the argument.

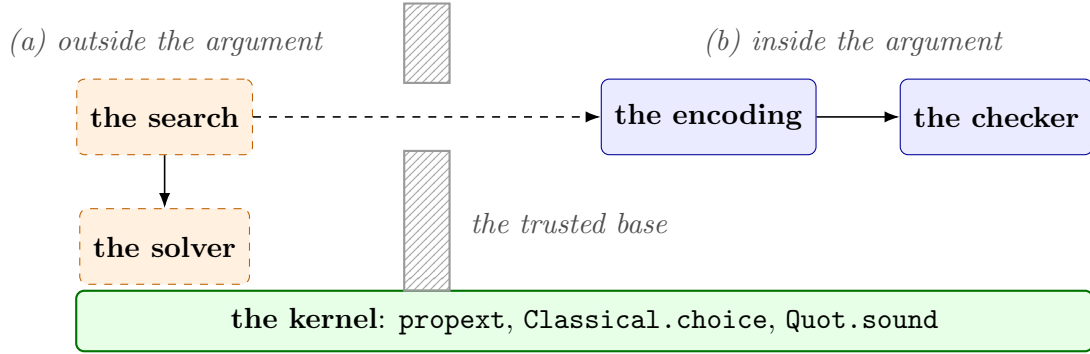


Figure 1: A distance claim crosses the wall as one object: the claim itself, and the file of steps that is supposed to certify it. The search and the solver may be as heuristic as they like, because everything beyond the wall is checked, and both rooms stand on the kernel.

2 What is in the library

QECCertificates is a Lean 4 development of 58 modules on Lean 4.34 [4], built against mathlib [5] at a pinned revision, on the same toolchain as the two developments that consume it.

- **A layer for the two-element field.** Vectors over that field, row spaces, a trusted row reduction with a proved pivot invariant, kernel bases, membership decisions that are computable, weight-limited enumeration with a covering theorem, and a certificate for a distance lower bound as a single decidable predicate.
- **An operator layer.** Pauli expressions with a symplectic representation, in which a Pauli operator is recorded by two bit-vectors per qubit saying whether it carries an X and whether it carries a Z, and translation theorems that make the vector-level decisions equal to statements about the matrices the surrounding ecosystem uses.
- **A certificate framework.** An in-kernel checker for LRAT proofs, the format in which a satisfiability solver records its refutation [6] – the descendant of the clausal proofs that DRAT-trim checks for the competitions [7] – with a proof that the checker is sound; a formalization

of the encoding step that ties satisfiability of the encoded formula to the existence of a light logical operator, in both directions; and symmetry breaking by lex-leader predicates, which keep the lexicographically smallest representative of each symmetry orbit, with the faithfulness of the reduction proved rather than assumed.

- **Instances.** Kernel-checked assertions of the triple (n, k, d) , the number of physical qubits, of logical qubits and the least weight of an undetected logical error, for a table of codes, including a hypergraph-product family covered by general theorems rather than by instance-by-instance evaluation; the shared instance families are Bacon-Shor [8], BB [9], hypergraph-product [10] and the lifted product [11].

3 What is checked, and how

Two properties are meant to be machine-checked rather than asserted, because both are the kind of claim that is either checked or is marketing. What a claim becomes is worth stating once, because the two shapes of a certified bound are not symmetric: the lower bound has to exclude every light operator and needs a solver for it, while the upper bound is one operator away.

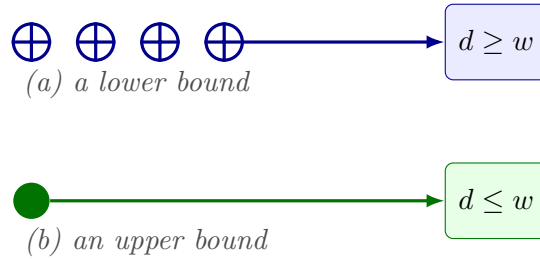


Figure 2: The two shapes of a certified bound, and why they are not symmetric. A lower bound in panel (a) has to rule out every logical operator lighter than the claimed weight w , which is why it needs a search and a solver. An upper bound in panel (b) is a single operator of weight w : write one down, have its weight checked, and the distance is at most w . The library spends its machinery on the left-hand side.

The trusted base. No declaration in the library uses `sorry`, a custom axiom, or `native_decide`; the computations that matter are carried out by the kernel itself, so a proof that leans on such a computation is a proof the kernel has already performed. The root module carries an audit region of 1,082 lines, each a request that the kernel print the axioms one load-bearing declaration depends on, and the region covers every non-private theorem and lemma in the package. A script in `tools/` reads those lines out of a build log and fails on any axiom outside `propext`, `Classical.choice` and `Quot.sound`; it fails as well when the log carries no output at all, so a clone that was never built is not a pass.

The build. Continuous integration builds the library on a hosted runner, and holds the result to zero error lines and zero warnings from this package’s own files. Not all of it fits. Eleven modules close their statements by kernel reduction over an object large enough to need tens of gigabytes, and the largest of them needs 77 GiB, so the job builds the 25 modules whose measured peak fits a runner, together with everything those modules import. A table in the repository records the peak of every module and fails the build when the tree and the table disagree, so a new module is

measured before it is built anywhere. The audit region is the root module, which imports the whole library, so the audit runs on a machine where the full build fits, and each run says in its log what it left out.

4 How the two companion developments use it

The library is the shared core of two further developments. One builds an interface layer in which the row space defined here is proved equal to the row space of Lean-QEC, the symplectic predicate is proved equivalent to commutation of actual Pauli matrices, and a state on the logical register is bounded by the logical dimension proved here. The other builds fault complexes, the gauge-fixed instances, and a layer of conditions under which a logical measurement succeeds, all on the same core. Both pin the package by commit, and the papers that report those results cite this library as the home of the shared statements, so a reader can check the interface claims without reading either paper's repository.

5 Using it

```
# in a lakefile.toml
[[require]]
name = "QECCertificates"
git = "https://github.com/QCL-SUAT/QECCertificates.git"
rev = "<40-hex commit>"

lake exe cache get    # prebuilt mathlib, on a machine that has not built it
lake build
```

Nothing is configured per machine: every dependency is fetched at a pinned revision, and a script that links a local prebuilt layer is optional and never required for a build to succeed. The released version [12] is archived at <https://doi.org/10.5281/zenodo.23056679>, which is what to cite. The audit region is part of the build, so a downstream user sees the axiom list for the statements they consume in their own build log.

6 What it does not do

The library does not simulate circuits, decode syndromes, or enumerate codes; those are served by mature projects, and the point of a certificate layer is to sit beside them rather than to reimplement them. It does not claim a general decision procedure for distances: what it offers is a format in which a bound, once found, is checkable, and a kernel that decides the check. It does not accept a solver's word at any step: the in-kernel path closes the argument with a proof object, and the certificate toolchain that lands next is written so that the checker is independent of the solver that produced the proof. Finally, it does not carry the code tables of the field: a handful of instances are included because they are load-bearing for the two companion developments, not as a database.

7 Conclusion

A distance claim can now be checked without repeating the search. The library supplies a proof object a third party can check, an audit region that prints the axioms behind every statement, and a

checker that shares no code with the solver. The Python certificate chain follows: the CNF encoder for distance lower bounds with a known-value gate, LRAT and RUP checkers written twice in different languages so that a bug in one is unlikely to be mirrored in the other, and the adapters that turn a check matrix into a certified bound. A later white paper will report that chain’s end-to-end results on the instances the two companion developments use.

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